Summing the charge sectors of the memory fluctuations
From central windows to the infinite-temperature state

Alan Day Chengheng*

Local research note — 6 November 2026

Abstract

We extend the project's conserved-subtracted memory coefficients to all charge sectors and control their charge tails. A uniform Hamiltonian variance bound and a positive regression denominator remove endpoint ambiguities. The central estimates extend to a logarithmic window, while a coarse polynomial bound suffices outside it. The full infinite-temperature spectral matrix equals the central one. The rescaled charge is Gaussian and factors from this two-energy matrix. This also supplies a precise energy-diagonal subtraction for the subsequent finite-memory problem.

1 The full state and the statement

Use the same overlapping complex Gaussian p-body Hamiltonian, standard CAR, $\hbar = 1$, string-time units, and common coupling variance

$$w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda_N = p^2/N \to \lambda > 0. \tag{1}$$

N is even, $\lambda_N \in [\lambda_-, \lambda_+] \subset (0, \infty)$, and $p \le N/4$ eventually. The annealed state at $\beta = \mu = 0$ is

$$\mathbb{E}\, 2^{-N} \operatorname{Tr} X = \sum_{m=0}^{N} b_N(m) \mathbb{E} \operatorname{tr}_m X_m, \qquad b_N(m) = 2^{-N} \binom{N}{m}, \quad x_m = (m - N/2)/p. \tag{2}$$

The charge decomposition is standard in complex SYK [1]; the present question is uniformity for the *subtracted* memory fields, whose raw coefficients can grow with p.

Let $\mathbf{V}_m = (Y_m, L_m, U_{2,m})$ be the actual first three collective memory coefficients with the conserved subtractions of the central-raw note [2]. Their all-sector definition and estimates are given below. If $d\boldsymbol{\nu}_*$ denotes the central positive two-energy matrix established there, then

$$\boxed{d\boldsymbol{\nu}_N^{\text{full}} = \sum_m b_N(m) d\boldsymbol{\nu}_{N,m} \implies d\boldsymbol{\nu}_*.} \tag{3}$$

Convergence is for bounded continuous energy tests, in particular fixed real-time correlators. A stronger joint statement is

$$d\boldsymbol{\nu}_N(x, E, E') \Longrightarrow \sqrt{\frac{2\lambda}{\pi}} e^{-2\lambda x^2}\, dx\, d\boldsymbol{\nu}_*(E, E'). \tag{4}$$

This is factorization of the indicated spectral matrix, not a claim of independence of the complete finite-N operator algebras.

*AI research assistant (OpenAI).

2 Endpoint sectors and the variance bound

Write $h = H_m$, $C_m = \mathbb{E}\,\mathrm{tr}_m\, h^2$. The exact variance is

$$C_m = w_N \binom{m}{p} \binom{N-m+p}{p}. \tag{5}$$

For $m < p$, $H_m = H_{m-1} = 0$ and the memory vanishes; all three fields and their counterterms are defined to be zero. The vacuum memory is likewise zero. For every $p \le m \le N$, $C_m > 0$. Term by term,

$$\begin{aligned} \frac{C_m}{J_s^2/\lambda_N} &= \prod_{j=0}^{p-1} \frac{4(m-j)(N-m+p-j)}{N^2} \\ &\le \prod_{j=0}^{p-1} \left(1 + \frac{p-2j}{N}\right)^2 \le e^{2p/N}. \end{aligned} \tag{6}$$

Indeed the difference of the two quadratic numerators is $(2m-N-p)^2$, and $\sum_{j=0}^{p-1}(p-2j) = p$. The same maximum is attained in the continuous charge variable at $m = (N+p)/2$, up to integer rounding. Denote the right-hand variance bound by M_N; it is bounded uniformly in the limit. Also, exactly on active sectors,

$$\frac{C_{m-1}}{C_m} = \frac{(m-p)(N-m+p+1)}{m(N-m+1)} \le p+1. \tag{7}$$

Set $A_m = \sum_i [H, c_i]^\dagger [H, c_i]$, $B_m = h^2 - C_m I$, and $O_m = \mathsf{W}_2 A_m$. The covariance eigenvalues are $\kappa_a = w_N \binom{m-a}{p} \binom{N-m+p-a}{p}$ on the supported rank subspaces. They lie in $[0, C_m]$. With their normalized weights ω_a, Wick contraction gives

$$\|B_m\|_2^2 = C_m^2(1+\chi_m), \qquad \chi_m = \sum_a \omega_a \kappa_a / C_m \in [0,1]. \tag{8}$$

Therefore $c_m = \langle O_m, B_m\rangle / \|B_m\|_2^2$ is well defined even at $m = p, N$. No continuation through a missing rank denominator is needed: use these invariant inner products.

We use normalized annealed Schatten norms below. Gaussian moments obey $\mathbb{E}\,\mathrm{tr}_m\, h^{2k} \le (2k-1)!! C_m^k$. One direct proof writes $h = \sum g_i X_i$, $\sum X_i^2 = C_m I$ and applies Gaussian integration by parts. The map $\Phi(X) = \sum X_i X X_i$ divided by C_m is unital and trace preserving, hence contractive on Schatten norms. Hölder bounds each of the $2k-1$ resulting terms by $C_m \mathbb{E}\,\mathrm{tr}_m\, |h|^{2k-2}$, completing the induction.

3 All-sector counterterms and a coarse global bound

For $m \geq p$ use the central-raw formulas without rank cutoffs: $a_m = \mathbb{E}\,\mathrm{tr}_m A_m$, $K_A = a_m I + c_m B_m$, and

$$\begin{aligned} P_m &= (ac_m + b)h^3 + [\bar{t}_m - (ac_m + b)\bar{z}_m]h, \\ \bar{t}_m &= \langle h, T_{3,m}\rangle / C_m, \qquad\qquad \bar{z}_m = \langle h, h^3\rangle / C_m. \end{aligned} \tag{9}$$

Here a, b are the fixed bounded functions of q_N, λ_N in Ref. [2]. Define $r = p/m$ and

$$\begin{aligned} C_{0,m} &= K_A - mr^2h^2, \\ C_{1,m} &= P_m + 2rhK_A - mr^3h^3, \\ D_{2,m} &= 2rhP_m + 3r^2h^2K_A - K_A^2/m - mr^4h^4, \\ C_{2,m} &= D_{2,m} + e_{\mathcal{Q},m}I + \delta h^4 + g_J h^2 K_A + b_R h P_m \\ &\quad + C_m[p\ell_Q + k_Q + 2\lambda_N x_m \ell_Q + o_Q c_m + 8\lambda_N q_N^2 c]B_m. \end{aligned} \tag{10}$$

All dimensionless coefficients and the raw polynomial $\mathcal{Q}$ are exactly those defined in Ref. [2]; $e_{\mathcal{Q},m} = \mathbb{E}\,\mathrm{tr}_m \mathcal{Q}_m$ is computed before any limit. These formulas are now a definition outside the central window, with no assertion of a sectorwise asymptotic expansion there. Every term commutes with h. For $\widehat{M}_m(s) = G_m + isZ_{1,m} - s^2 Z_{2,m}/2 + O(s^3)$ put

$$Y_m = mG_m - C_{0,m}, \quad L_m = mZ_{1,m} - C_{1,m}, \quad U_{2,m} = mZ_{2,m} - C_{2,m}. \tag{11}$$

The normalized map $\mathrm{Rem}\,/m$, where $\mathrm{Rem}\,X = \sum c_i^\dagger X c_i$, is unital and preserves normalized traces from $m-1$ to m. It is Schatten contractive. Thus the exact identity $A_m = \mathrm{Rem}\,H_{m-1}^2 + (2p-m)h^2$ gives $\|A_m\|_{2k} \leq K_k N M_N$. Equations (7)–(8) also give $|c_m| \leq KN^2$ and $\|K_A\|_{2k} \leq K_k N^2 M_N$. The explicit three-contraction formula for $\bar{t}_m$ gives $|\bar{t}_m| \leq KN^2 C_m$, while $|\bar{z}_m| \leq 3C_m$. To see the first bound without small-variance division, each binomial covariance ratio in t_a/C_m is at most $p+1$ and its reduced coefficient α_a lies in $[0,p]$. Consequently $\|P_m\|_{2k} \leq K_k N^2 M_N^{3/2}$.

All raw quartic words in $\mathcal{Q}$ are bounded by $K_k N M_N^2$, using Rem, Gaussian moments and Hölder. The exact renewal formula for mZ_2 has only these words, products $hT_3, h^2A, A^2/m, h^4$, and coefficients bounded by powers of N with $r \leq 1$. In (10) the largest crude power is K_A^2/m. Therefore, with constants independent of m,

$$\|Y_m\|_{2k} \leq K_k N^2 M_N, \quad \|L_m\|_{2k} \leq K_k N^2 M_N^{3/2}, \quad \|U_{2,m}\|_{2k} \leq K_k N^4 M_N^2. \tag{12}$$

These deliberately coarse bounds are used only on exponentially rare charge sectors, never to estimate the central fluctuations.

4 Auditing a logarithmically growing central window

Fix any $A_0 > 0$ and set $R_N = \sqrt{A_0 \log p}$. The existing central calculation extends uniformly to $|x_m| \le R_N$. The needed consequence is

$$\sup_{|x_m| \le R_N} (\|Y_m\|_2^2 + \|L_m\|_2^2 + \|U_{2,m}\|_2^2) \le K_{A_0}. \tag{13}$$

This is not inferred from compact-window uniformity alone. The remainder audit is as follows.

Rational filters and the sixth moment. All scaled denominators of the exact filters stay separated from zero, since $x_m/p = O(\sqrt{\log p}/p)$ and $a, b \le p, r \le 2p$. The finite sixth-moment recursion has rational coefficients in $\epsilon = N^{-1/2}, A, B, z = x_m\sqrt{\lambda_N}$. The 21 negative-power cancellations are polynomial identities in z, and its leading six moments are independent of z. After cancellation, fixed-order Taylor remainders are bounded by $K\epsilon(1+|z|)^D$ for a fixed finite D: each denominator is a finite product of factors $1 \pm 2z\epsilon + O(\epsilon^2)$. This gives a uniform sixth-moment bound on the logarithmic window.

The coherent force and input tails. In the exact coherent norm series, truncate input deficits u, v and the Racah index j at $\lceil \log p \rceil$. The input and summand ratios have fixed factorial envelopes $K^u/u!$ and $K^j/j!$ throughout this window of charges. The discarded terms, even multiplied by the old p^4 crude bound, vanish faster than every power of p. On the retained set the logarithmic expansion has a remainder p^{-3} times a fixed polynomial in u, v, j, x_m. The exact p^2, p cancellations and the x_m-independence of the finite term therefore leave $p^{-1}(\log p)^D$. This proves the central weighted A norm bound here as well.

Strong substitutions. The A-filter error uses the sixth moment and an output cutoff $S \le p^{3/4}$ as before. The H^2 filters subtract their exact $p\ell$ and $2\lambda_N x_m \ell$ terms first. On $S \le p^b$, their Taylor remainder is p^{-1} times a fixed polynomial; choose fixed $b > 0$ small enough for this polynomial degree. On the complement, the linear filter bound and sixth moment give an L^2 tail tending to zero. All extra charge factors are powers of $\log p$. The cubic rank slope, centered B_S, B_Z, and the F proxy have the same factorial-weighted estimates. The general force identity has errors proportional to x_m/p. It replaces $\mathcal{S}_N$ by $(O_- + O_+)/2$ without a new charge transport argument. Finite Gram projections are stable because their limiting Gram is strictly positive at fixed λ.

Marked energy factors. The balanced-occupation trace proof gives error at most $p^{-1}e^{KR_N}(1+R_N)^D$ per fixed marked graph after averaging over overlaps; the same exponential overlap moments bound its tail. This is $p^{-1+o(1)}$. It bounds all fixed energy insertions of the marked representatives. The invariant Gaussian moment bound controls multiplication of the linear filter errors by h. Thus the restored raw expansion has a uniformly bounded leading part and a vanishing strong remainder, proving (13).

5 Charge tails, full spectra, and joint factorization

For $m \sim \mathrm{Bin}(N, 1/2)$ the elementary Chernoff bound is

$$\Pr(|x_m| > R) \le 2e^{-2\lambda_N R^2}. \tag{14}$$

For example it follows from $\mathbb{E}e^{t(m-N/2)} = \cosh(t/2)^N \le e^{Nt^2/8}$ and optimization over t. Choose $A_0 = 20/\lambda_-$ in the logarithmic cutoff. The squared version of (12) gives

$$\sum_{|x_m|>R_N} b_N(m)\|\mathbf{V}_m\|_2^2 \le KN^8 e^{-2\lambda_- A_0 \log p} \le K' p^{-24} \longrightarrow 0. \tag{15}$$

Here $\|\mathbf{V}\|_2^2$ is the sum of the three norm squares; M_N is absorbed into the constant. For a fixed $L < R_N$, (13) controls the intervening annulus:

$$\sum_{L<|x_m|\le R_N} b_N(m)\|\mathbf{V}_m\|_2^2 \le 2K_{A_0} e^{-2\lambda_- L^2}. \tag{16}$$

Together these establish the required uniform integrability:

$$\lim_{L\to\infty} \limsup_{N\to\infty} \sum_{|x_m|>L} b_N(m)\|\mathbf{V}_m\|_2^2 = 0. \tag{17}$$

For fields V_i, V_j, a bounded two-energy test has matrix element bounded in modulus by its supremum norm times $\|V_i\|_2\|V_j\|_2$. Thus (17) controls both diagonal and cross measures, including their total variation in the charge tails. On $|x_m| \le L$ use the already proved compact-window spectral limit, independent of x_m. First let $N \to \infty$ there, then $L \to \infty$. This proves (3); no sum over energy-diagonal weights or oscillatory phases was discarded.

The charge characteristic function is exactly

$$\mathbb{E}e^{itx_m} = \cos\left(\frac{t}{2p}\right)^N, \qquad \log \mathbb{E}e^{itx_m} = -\frac{t^2}{8\lambda_N} - \frac{t^4}{192\lambda_N p^2} + O(p^{-4}). \tag{18}$$

It converges to the centered Gaussian with variance $1/(4\lambda)$. For a bounded continuous charge test $f(x)$ and bounded continuous energy test $g(E, E')$, the same compact-window argument gives

$$\sum_m b_N(m) f(x_m) \int g\, d\boldsymbol{\nu}_{N,m} \longrightarrow \left[\int f(x)\sqrt{\frac{2\lambda}{\pi}} e^{-2\lambda x^2} dx\right] \int g\, d\boldsymbol{\nu}_*. \tag{19}$$

Tightness and approximation on compact sets yield (4). The charge fluctuation therefore does not add a new leading dynamic spectral component once the correct sector-dependent conserved polynomials have been removed. The ordered small-λ results from the previous raw-spectrum note transfer to this full state after the fixed-λ double-scaled limit.

6 A definite finite-memory subtraction and executed evidence

For each sample and charge let

$$\mathsf{P}_h(X) = \sum_\epsilon P_\epsilon X P_\epsilon, \qquad h = \sum_\epsilon \epsilon P_\epsilon. \tag{20}$$

This is the Hilbert–Schmidt orthogonal projection onto the operators commuting with h, including degeneracies. It equals the infinite symmetric time average of $e^{ith}Xe^{-ith}$ at finite dimension. Define the actual finite-memory dynamic field

$$\mathfrak{F}_{N,m}(s) = (1-\mathsf{P}_h)[m\widehat{M}_m(s)], \qquad \mathfrak{F}_{N,m} = 0 \quad (m < p). \tag{21}$$

This subtracts the complete samplewise energy-diagonal part. Its finite-N Taylor coefficients through order two are exactly $(1-\mathsf{P}_h)(Y, L, U_2)$, because all $C_{j,m}$ commute with h. Moreover

$$\|\mathsf{P}_h V\|^2_{2,\mathrm{full}} = \nu^{\mathrm{full}}_{V,N}\{E = E'\} \longrightarrow 0, \quad V = Y, L, U_2. \tag{22}$$

The limiting positive diagonal measures in (3) are absolutely continuous for $0 < q < 1$. Their mass on the closed energy diagonal is zero, so weak convergence and the closed-set inequality prove (22). Consequently this exact subtraction preserves the previously calculated limiting Taylor-coefficient matrix. It specifies the next problem without guessing a finite-s polynomial background.

What remains is to bound and compute, for fixed real s, u, T, $\mathbb{E}\,\mathrm{tr}\,\mathfrak{F}_N(s)^\dagger e^{iTH}\mathfrak{F}_N(u)e^{-iTH}$. The three known derivatives do not bound a finite-memory Taylor remainder. The projection involves spectral information of each sample and does not itself establish a local bulk reconstruction.

Checks. Exact binomial/product comparisons verify 6,766 sector variance and adjacent-charge cases. Independent Gaussian-integer CAR contractions cover every charge in the three systems $(N, p) = (3, 1)$, $(4, 2)$ and $(5, 2)$: 15 sectors, including inactive and fully filled ones. They verify the denominator, A mean, B–O cross and regression residual. Twelve new full-rank growing-window groups use $x = \pm\lceil\sqrt{4\log p}\rceil$, $\lambda = 0.25, 0.5$, $p = 32, 64, 128$, at 80 and 120 digits. Four positive binomial-tail sums check the outer cutoff. Six prior/current certificates and their source hashes were verified; old scans were not rerun. Finite-size differences and cancellation factors remain in the JSON records. These numerical refinements are not interval bounds; the logarithmic-window and summation proofs are the preceding sections.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Secs. 4.3 and 5. Charge projection and existing ensemble structure.

[2] A. Day and Chengheng, local project notes: *The raw memory coefficient in a central charge window*; *Coherent memory norms on central charge windows*; *Central charge windows for the memory calculation*; *Central force spectrum* (6 November 2026).